

YI ZHU

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School of Architecture and Urban Planning, Nanjing University, Nanjing, China

EDUCATION

Master of Engineering, Nanjing University, China (Sep. 2024 – Jun. 2027, expected)
Department of Urban Planning, School of Architecture and Urban Planning **GPA: 90.65/100**
Thesis Title: Research on Land Use Mix and Application of Large Language Models
Supervisor: Prof. Xiao Qin, Prof. Shanqi Zhang, Prof. Feng Zhen

Bachelor of Engineering, Soochow University, China (Sep. 2019 – Jun. 2024)
Department of Urban Planning, School of Architecture **GPA: 3.5/4.0**
Supervisor: Prof. Fei Yu and Prof. Yasi Tian

School of Medicine, Soochow University, China (Sep. 2019 – Jun. 2020)

RESEARCH INTERESTS

- Health and feminist geography: activity-based exposure, spatial inequality, and gendered mobility
- Urban built environment: land use mix, urban renewal, and planning history
- GeoAI and computational methods: large language models, knowledge graphs, and spatial analysis

JOURNAL PUBLICATIONS

1. **Yi Zhu**, Xiao Qin, Shanqi Zhang, Yuanyuan Xu, Ruotong Wu, Feng Zhen. Mixed land use without economic absorptive capacity exacerbates multidimensional inequality. *npj Urban Sustainability*, peer review.
2. **Yi Zhu**, Xiao Qin, Ruotong Wu, Feng Zhen. How is commute mode choice related to 2D/3D land use mix? A Case Study of Central Suqian City. *Transportmetrica B: Transport Dynamics*. <https://doi.org/10.1080/21680566.2026.2622498>
3. **Yi Zhu**, Yasi Tian, Guoyan Tang, Dantong Zheng, Fei Yu. Spatial Patterns and Influencing Factors of People's Commune Sites: A Case Study of Henan Province, China. *Land*. <https://doi.org/10.3390/land13111860>
4. Fei Yu, **Yi Zhu**, Lu Ye. Design, Evolution, and Characteristics of the Residential Buildings of People's Commune (1958-1983). *Journal of Architecture*. <https://doi.org/10.1080/13602365.2026.2628019>.
5. Fei Yu, **Yi Zhu**, Xukai Zhao, Xiao Qin. Rural Heritage under Socialism: Analyzing Distribution, Characteristics, and Impact of China's People's Communes with Large Language Models and Knowledge Graphs. *Journal of Asian Architecture and Building Engineering*, revisions for first review.
6. Yuneng Jiang, **Yi Zhu**, Yasi Tian. Measuring the Urban Sprawl of a Mega-Urban Agglomeration Area Based on Multi-Dimensions with a Mechanical Equilibrium Model: A Case Study of the Yangtze River Delta, China. *Land*. <https://doi.org/10.3390/land12081548>
7. Fei Yu, **Yi Zhu**, Dantong Zheng. Revitalizing Township Industrial Heritage: Adaptive Reuse Strategies Based on the Practice of Jiangsu Jinshiyuan Liquor Industry Co., Ltd. *Huazhong Architecture* (in Chinese) <https://doi.org/10.13942/j.cnki.hzjz.2024.10.027>

CONFERENCE PAPERS OR PRESENTATIONS

1. **Yi Zhu** (Aug. 2026). Heritage Characteristics Interpretation and Cluster Identification of Rural Collectivization Period Sites Based on Knowledge Graphs and Machine Learning. *2026 Annual National Planning Conference*, Guangzhou, China.
2. **Yi Zhu**, Xiao Qin (Jul. 2026). Impacts of land use mix on inequality. *The 20th International Association for China Planning (IACP) Conference*, Xi'an, China. (Presentation)

3. **Yi Zhu**, Xiao Qin, Ruotong Wu, Feng Zhen (Nov. 2025). Impacts of 2D/3D land use mix on residents' commuting behavior: A case study of central Suqian city. *2025 Annual Conference on Territorial and Spatial Planning of China Society of Natural Resources*, Xiamen, China.

RESEARCH PROJECTS

1. **Project Leader**, 'Mining Socio-Emotional Perceptions in Urban Renewal Driven by Multi-Agent Large Language Models', funded by Jiangsu Provincial Postgraduate Research and Practice Innovation Program (Natural Science Category), 2026–2027.
Independently established the research framework and took full responsibility for project design, data collection, analysis, and reporting.
2. **Research Member**, 'Intelligent Identification and Optimization of Inefficient Urban Land Use Based on Element Flow Simulation', funded by National Natural Science Foundation of China, 2024–2028.
Conducted a systematic literature review on inefficient land use and contributed to the development of evaluation indicators for inefficient land identification.
3. **Research Member**, 'Research on the Spatial Production Mechanism and Heritage Value of Rural People's Communes from the Perspective of "Sphere": A Case Study of Henan Province', funded by National Natural Science Foundation of China, 2024–2028.
Participated in heritage field surveys and applied GIS-based spatial analysis to the study of People's Commune heritage sites.
4. **Project Leader**, 'Urban Vitality Assessment and Optimization Based on Multi-Source Data and Multi-Dimensional Evaluation', funded by the 23rd Undergraduate Extracurricular Academic Research Fund, Soochow University, 2021–2022.
Independently led the full research process, including research design, data collection, spatial analysis, and manuscript writing.
5. **Research Member**, 'Performance Evaluation and Optimization of Urban-Rural Infrastructure Integration in Jiangsu Province from the Perspective of Urban-Rural Integration', funded by the National College Student Innovation and Entrepreneurship Program, 2022–2024.
Responsible for literature review, data collection, and participated in subsequent manuscript writing.

SELECTED INDUSTRIAL PROJECTS

1. **Core Member**, 'Research on Jobs-Housing Balance in Suqian City', 2025.
Participated in the full project workflow, including field investigation, data processing, problem diagnosis, and planning strategy formulation.
2. **Core Member**, 'The 15th Five-Year Plan of Nanjing Metropolitan Circle', 2026.
Responsible for early-stage data collection, including the compilation and analysis of inter-firm linkage data across the Nanjing Metropolitan Circle.

HONORS, MEMBERSHIPS & SERVICE

Honors & Awards

1. First Prize for Outstanding Graduate Presentation, 2025 Annual Conference on Territorial and Spatial Planning of China Society of Natural Resources. 2025.
2. Second-Class Academic Scholarship, Nanjing University. 2024 & 2025.
3. Second Prize, National College Student Natural Resources Competition, China. 2023.

Professional Memberships

- Student Member, Architectural Society of China (ASC). 2024–Present.

Academic Service

- Peer Reviewer, *Cities* (Elsevier), 2026.
- Peer Reviewer, *npj Heritage Science* (Nature Portfolio), 2026.

Voluntary Service

- Volunteer Docent, Nanjing Urban Planning Exhibition Hall, Nanjing, China. Sep. 2025–Jul. 2026.
- Online Teaching Volunteer, provided mathematics tutoring (Grade 8 level) to left-behind children in rural areas. 2023.

SKILLS

- **Geospatial Analysis:** ArcGIS, GeoPandas, OSMnx; spatial regression, accessibility modeling, network analysis, and multi-scale spatial data integration for urban planning and health geography research.
- **Computational & Statistical Methods:** Python (pandas, scikit-learn, TensorFlow); BERT fine-tuning, LLM API integration, knowledge graph construction (Neo4j), XGBoost/SHAP, and multi-agent workflow design; Stata, SPSS for quantitative analysis.
- **Design & Visualization:** AutoCAD, Rhino Grasshopper, Photoshop; urban planning drawing and graphic design.
- **Languages:** Chinese (native), English (CET-6), Japanese (JLPT N5).